

Computer Networks Mini Project

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System Architecture and Flow Design

1. Architectural Overview

This system utilizes **Software-Defined Networking** (SDN) to fundamentally separate the network's data and control planes. By unbundling these functions, the architecture transitions from traditional, vertically integrated routers to a programmable, layered network model.

The system operates across three distinct tiers:

- **Data Plane:** This layer consists of the network's switches, which are relatively simple, fast devices designed to execute "match plus action" rules stored in their flow tables.
- **Control Plane:** This logically centralized layer consists of servers and software that determine, manage, and install the flow tables within the data-plane switches.
- **Application Layer:** Network-control applications act as the "brains" of the SDN control plane, using APIs provided by the controller to specify how the data plane devices should route specific traffic.

2. Component Specifications

Our specific emergency notification implementation maps to the SDN architecture using the following components and interfaces:

- **Network-Control Application:** A custom routing application that identifies emergency UDP broadcasts (e.g., on port 12000) and dictates simultaneous forwarding to registered subscriber hosts.
- **Northbound API:** The interface that allows the network-control application to communicate with the Ryu controller, accessing network state information and pushing down routing intents.
- **SDN Controller (Ryu):** The network operating system that maintains accurate network state information and translates the application's requirements into low-level switch commands.
- **Southbound API (OpenFlow):** The communication protocol used between the Ryu controller and the Mininet switches. The controller uses this protocol to push flow table updates to the switches.
- **Data Plane (Open vSwitch in Mininet):** The virtual switches that perform the hardware-level matching of incoming UDP packets against the flow table, forwarding matched emergency traffic with high priority or querying the controller for unmatched traffic.

3. System Flow Diagrams

Diagram 1: SDN Architectural Stack Separation

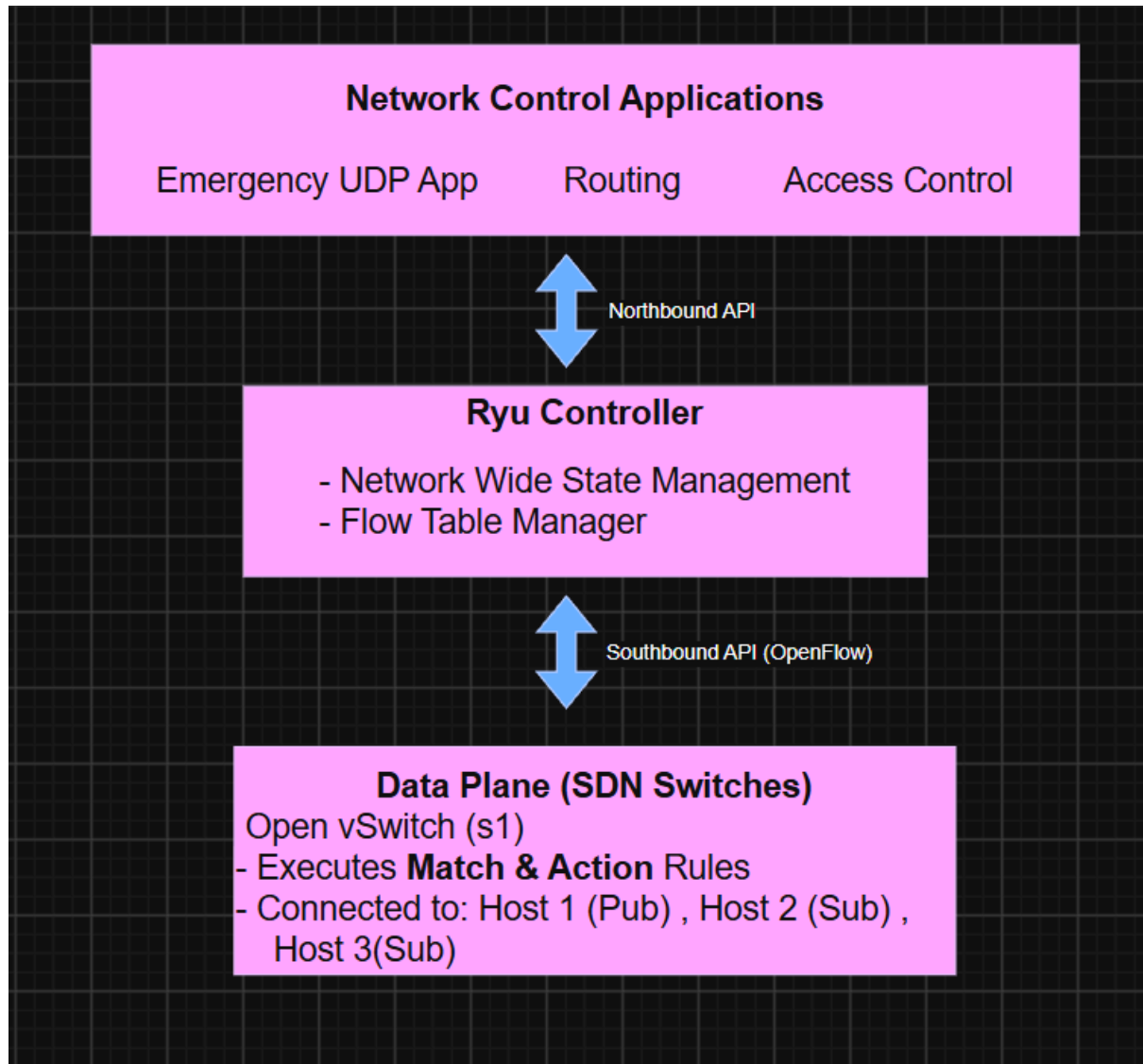


Diagram 2: Emergency UDP Packet Flow & Control Plane Interaction

